Wavelets and Autoencoders Are All You Need

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Abstract

N-BEATS [Oreshkin et al., 2020] broke new ground in univariate time series forecasting by using fixed basis functions in a neural basis expansion in which the network learns coefficients for a frozen basis function and then expands those coefficients into the desired forecast and backcast lengths through a linear transformation. Their innovation came at an extreme cost, as a typical N-BEATS model can range from 25M–44M parameters. In this paper we extend the idea of solving for coefficients of a frozen basis function but, instead of relying on a global Fourier function, we use wavelets, which provide localization in that they can identify and locate components of the global signal — a key requirement in time series analysis algorithms. Wavelets have been used extensively in signal analysis for feature extraction, denoising, and multi-resolution analysis. Likewise, autoencoders are useful constructs for identifying key characteristics of a signal while filtering out non-essential transient components, and are useful in domains such as denoising signals and stock trading. We introduce a variety of novel block types that use a combination of discrete wavelet transforms (DWT) and autoencoders. Our novel architecture improvements reduce parameter counts by 60–95% over N-BEATS while surpassing or equaling prior published results without ensembling. In addition, we introduce a family of combined TrendWavelet block types that combine a low-pass trend basis with a high-pass wavelet basis to produce <1M parameter models that score within 0.5% sMAPE of 19M–43M parameter baseline N-BEATS models, representing a 12–80× compression with little to no loss of accuracy. Furthermore, we show that these new block constructs are transferable to other architectures like NHiTS and Transformers with equally impressive reductions in parameter counts and competitive accuracies.

1 Introduction and Prior Work

1.1 Introduction

Time series forecasting and signal analysis are two names for the same thing, even though practitioners in each discipline use their own terminologies: moving average vs. finite impulse response (FIR) filter, autoregressive filter vs. infinite impulse response (IIR) filter, seasonal decomposition vs. band-pass filter of seasonal frequencies. A time series dataset (sales figures, temperature, stock prices) is just a discrete set of sampled data. Likewise, all deep learning problems can be reconsidered as signal analysis problems. For example, image recognition in deep learning is often performed by sampling an image and passing the samples through a convolutional filter, similar in principle to what one might do in Photoshop when applying a sharpening filter. Deep learning, then, can be seen as a type of probabilistic signal analysis. Why, then, don't we use more techniques developed in signal analysis in machine learning?

Signal analysis tools like the Discrete Fourier Transform and the Discrete Wavelet Transform were developed as techniques used in signal processing, migrated into forecasting, and are trickling into

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

machine learning. N-BEATS was a pioneer of sorts in bringing some traditional signal analysis tools into AI with its use of basis expansion blocks. The N-BEATS Seasonality and Trend blocks are literally signal-decomposition operators applied inside a neural network. The novelty was using the neural network to learn which decomposition coefficients best reconstruct the observed signal.

1.2 Wavelets

Fourier analysis decomposes a signal into a sum of sine and cosine waves of fixed frequency. It is a useful tool for analyzing stationary signals whose frequency content does not change over time. Real-world signals routinely break that mold, however. For example, a stock price may trend smoothly following the normal sales cycle and then spike abruptly because of a news article, a comment made by a prominent person in the news, or when a large hedge fund liquidates a considerable position. Fourier bases represent these transient features poorly and tend to smear localized events across the other frequency components of the signal.

Wavelets address this problem by providing basis functions that are localized in both time and frequency. Where a sine wave oscillates forever, a wavelet is a brief oscillatory burst with exponentially decaying tails and a zero mean. They oscillate in a particular pattern defined by their mother wavelet and then decay to zero.

The mother wavelet $\psi(t)$ is scaled and translated to produce a family of basis functions:

$$\psi_{j,k}(t) = 2^{j/2}\,\psi(2^j t - k)$$

where j controls scale (frequency resolution) and k controls translation (time position). Short, high-frequency wavelets have narrow time resolution; long, low-frequency wavelets have narrow frequency resolution. This makes wavelets substantially more effective for real-world time series analysis, where variance and frequency content change over time.

A discrete wavelet transform (DWT) brings the continuous wavelet into the digital domain. DWTs compute the inner product of a signal with a basis family, producing a set of orthonormal coefficients that can then be used to reconstruct the original signal [Mallat, 1989, Daubechies, 1992].

Wavelets have been used sparsely in machine learning. WaveTS [Zhou et al., 2025] learns a filter-bank specialized to fit non-stationary signals by learning the basis function through decomposing its n-th order derivatives and then learning an affine transformation that lets the network suppress and cross-mix frequencies in training.

MultiWave [Deznabi and Fiterau, 2023] is a model-agnostic preprocessor around any neural network architecture. It takes the concept of machine-learning/signal-processing equivalency to heart by preprocessing input data with a DWT, grouping the outputs by frequency, and then feeding the groups into any neural network (Transformer, FFN, CNN, etc.), feeding different groups into different networks. The networks' outputs are combined and an inverse DWT is applied to obtain the prediction.

W-Transformers [Sasal et al., 2022] uses a similar approach by preprocessing time-series data with a discrete overlapping wavelet transform and then feeding the results into a collection of transformer models to analyze the decomposed signals independently.

Most directly related, Pramanick et al. [2024] apply DWT preprocessing to N-BEATS itself, training separate Generic stacks per sub-band; our work differs by embedding the wavelet basis inside each block rather than as an external preprocessor.

1.3 Autoencoders

An autoencoder is a neural network trained to reproduce its input. It has an encoder that maps an input x into a smaller-dimensional latent code z, and a decoder that maps z back into a reconstructed version of the original input $\hat{x}$:

```
x -> encoder -> z -> decoder -> x_hat
loss = reconstruction_error(x, x_hat) + optional_regularization
```

The premise is that the model is constrained by the reduced dimensionality of the latent space. Therefore, since the decoder network does not have access to the original signal, a perfect copy is difficult. Thus the network is pushed to learn how to reconstruct the original signal from the information it can

extract from the latent representation. The classic paper of Hinton and Salakhutdinov [Hinton and Salakhutdinov, 2006] showed that autoencoders generalize PCA to non-linear domains. Autoencoders are also used widely for self-supervised pretraining, generative modeling, and image denoising and generation [Vincent et al., 2008].

2 Model Description

N-BEATS [Oreshkin et al., 2020] analyzes a time series through a sequential stack of doubly residual blocks, each contributing a backcast $\hat{\mathbf{x}}_\ell$ and forecast $\hat{\mathbf{y}}_\ell$ via a learned linear mapping from a shared 4-layer fully connected backbone:

$$\mathbf{h}_\ell = f_{\text{bb}}(\mathbf{x}_\ell^{\text{in}}), \quad \hat{\mathbf{x}}_\ell = g_b(\boldsymbol{\theta}^b), \quad \hat{\mathbf{y}}_\ell = g_f(\boldsymbol{\theta}^f)$$

where $\mathbf{x}_\ell^{\text{in}} = \mathbf{x} - \sum_{k<\ell} \hat{\mathbf{x}}_k$ is the residual backcast input to block ℓ, and g_b, g_f are basis expansion generators (defined explicitly in Section 2.3).

All N-BEATS blocks accept a residual backcast (or input if first block) $\mathbf{x} \in \mathbb{R}^L$ and return $(\hat{\mathbf{x}}, \hat{\mathbf{y}}) \in \mathbb{R}^L \times \mathbb{R}^H$.

2.1 Backbone Blocks

2.1.1 RootBlock

The legacy RootBlock is a 4-layer, uniform-width FFN with ReLU activation σ:

$$\mathbf{h} = \sigma(\mathbf{W}_4\, \sigma(\mathbf{W}_3\, \sigma(\mathbf{W}_2\, \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3) + \mathbf{b}_4)\,, \quad \mathbf{W}_i \in \mathbb{R}^{U \times U}$$

No bottleneck is imposed; all layers share trunk width U.

2.1.2 AERootBlock

The AERootBlock is a drop-in replacement for the legacy RootBlock backbone. It introduces an hourglass-shaped autoencoder that routes all information through a low-dimensional latent bottleneck $z \in \mathbb{R}^d$:

$$\mathbf{h} = f_{\text{dec}}(f_{\text{enc}}(\mathbf{x})), \quad f_{\text{enc}} : \mathbb{R}^L \to \mathbb{R}^{U/2} \to \mathbb{R}^d, \quad f_{\text{dec}} : \mathbb{R}^d \to \mathbb{R}^{U/2} \to \mathbb{R}^U$$

The bottleneck acts as a signal denoiser, regularizer, and feature extractor. The decoder must reconstruct both output paths from z, forcing the backbone to discard irrelevant information and retain the most informative features.

The parameter count of the backbone reduces to $O(UL/2 + Ud/2 + Ud/2 + U^2/2)$. At $U = 512$, $L = 480$, $d = 16$, this is approximately 147K — a 5.5× reduction relative to RootBlock at equivalent U. The interface to any downstream projection head remains identical: both backbones produce $h \in \mathbb{R}^U$.

2.1.3 AERootBlockLG

The LG variant of AERootBlock introduces a learnable sigmoid vector and scales the interior latent representation element-wise:

$$z_{\text{gated}} = \sigma(\gamma) \odot z$$

Because $\sigma(\gamma_i)$ saturates at zero or one at its limits, the network can shut off unnecessary latent dimensions, effectively performing a soft band-pass filter. This is particularly useful when the optimal bottleneck size is unknown: one can select a lower latent dimension because the network only uses the dimensions it needs. In testing, this manifests as networks using AERootBlocks needing a latent dimension $d = 32$ for the M4 dataset, but AELG-based networks needing only $d = 16$. For example, a 10-stack TrendWavelet block with AELG root and a latent dimension of 32 using a Daubechies-3 wavelet (`TWAELG_10s_ld32_db3`) uses only 0.48M total parameters while remaining competitive with 15M+ RootBlock models.

Table 1: Backbone architecture comparison.

Backbone	Layer topology	Latent operation	Extra params
RootBlock	input $\to U \to U \to U \to U$	None	Baseline
AERootBlock	input $\to U/2 \to d \to U/2 \to U$	Deterministic pass-through	Negligible
AERootBlockLG	Same as AE	$z \cdot \sigma(\gamma)$	d (gate vector)

Table 2: Backbone parameter counts at default widths ($L = 480$).

Backbone	Width U	Latent d	Backbone params	vs. RootBlock
RootBlock (Generic)	512	—	∼810K	1.0×
AERootBlock (Generic)	512	32	∼163K	5.0× fewer
AERootBlock (Generic)	512	16	∼147K	5.5× fewer
AERootBlockLG (Generic)	512	16	∼147K + 16	≈5.5× fewer
RootBlock (Trend)	256	—	∼319K	1.0× (Trend)
AERootBlock (Trend)	256	16	∼46K	6.9× fewer
AERootBlockLG (Trend)	256	16	∼46K + 16	≈6.9× fewer

2.1.4 Parameter efficiency

The standard width parameters (`g_width`=512, `s_width`=2048, `t_width`=256, `ae_width`=512) govern the main trunk width U for each block family. The `latent_dim` d is an additional hyperparameter that controls only the AE bottleneck and does not change the output dimension $\mathbf{h} \in \mathbb{R}^U$ seen by downstream projection heads.

The backbone parameter count formula for each family is:

- **RootBlock**: $UL + 3U^2$ — four uniform-width layers, no bottleneck.
- **AERootBlock**: $UL/2 + Ud + U^2/2$ — parameters scale roughly as $O(U \cdot d)$ at the bottleneck, which is minor compared to the main width cost.
- **AERootBlockLG**: same as AERootBlock plus d gate parameters — negligible overhead; practical benefit is that d can be halved relative to AERootBlock without accuracy loss.

Because `latent_dim` is typically 5–32, AE variants achieve comparable forecast accuracy with dramatically fewer total backbone parameters than their RootBlock counterparts when `latent_dim` is small and width is constrained. Table 2 gives concrete counts at default widths and a representative backcast length of $L = 480$.

The compression ratio grows with width: wider trunks benefit more from the hourglass bottleneck because the three large $U \times U$ layers in RootBlock scale quadratically while the AE bottleneck scales linearly in d. At Generic/Wavelet width ($U = 512$, $d = 16$) the ratio reaches 5.5×; at Trend width ($U = 256$, $d = 16$) it reaches 6.9×. This makes AE backbones attractive for TrendWavelet and TrendWaveletGeneric configurations, where the Trend trunk width is narrow by design and every saved parameter can be reinvested in stack depth or wavelet family selection, which improves expressivity of the model.

2.2 Branch-specific encoder-decoder designs

The backbone variants described above produce a single shared hidden representation $\mathbf{h}$ that is then split by separate linear heads into backcast and forecast coefficient vectors. A complementary design gives each output branch its own dedicated encoder-decoder sub-network. The `AutoEncoderAE` and `GenericAEBackcastAE` blocks follow this pattern: after the shared AE trunk compresses the input into a trunk latent code z, two independent hourglass sub-branches — one for the backcast path and one for the forecast path — each re-encode z through their own bottleneck before projecting to the output length.

$$z \xrightarrow{\text{enc}_b} z_b \xrightarrow{\text{dec}_b} \hat{\mathbf{x}}, \qquad z \xrightarrow{\text{enc}_f} z_f \xrightarrow{\text{dec}_f} \hat{\mathbf{y}}$$

This gives the network two independent compression points per block. The trunk bottleneck enforces a shared representation of the input window; the branch bottlenecks then clean up each signal

Table 3: Single-trunk vs. branch-specific AE designs.

Block	Shared trunk	Backcast branch	Forecast branch	`thetas_dim` role
`GenericAE`	AERootBlock	shared $\mathbf{h} \to$ linear	shared $\mathbf{h} \to$ linear	unused
`AutoEncoderAE`	AERootBlock	$z \to$ enc-dec $\to$ output	$z \to$ enc-dec $\to$ output	unused
`GenericAEBackcastAE`	AERootBlock	$z \to$ enc-dec $\to$ output	$z \to$ enc-dec $\to \boldsymbol{\theta}^f \to$ basis	coefficient projection

independently. The two objectives are separated downstream again to impart more diversity into the learned parameters.

The `latent_dim` hyperparameter controls the branch-local bottleneck size in these blocks, while `thetas_dim` remains reserved for explicit coefficient projections such as the `GenericAEBackcast*` forecast head. `AutoEncoderAE` omits `thetas_dim` entirely and lets both branch decoders project directly to the target lengths.

The branch-specific design adds approximately $4 \cdot U_b \cdot d_b$ parameters per branch, where U_b is the branch width and d_b the branch latent size. For typical settings ($U_b = U/2$, $d_b = d$), this doubles the AE parameter budget relative to a shared-head block, but keeps the total well below the RootBlock baseline since both the trunk and the branches remain in the hourglass regime.

2.3 Wavelet Family Blocks

The legacy N-BEATS basis-expansion blocks predict coefficient vectors and expand them through a frozen basis into backcast and forecast outputs matching the target lengths. The original interpretable version uses polynomial trend and Fourier seasonality bases. The wavelet blocks replace the Fourier seasonality basis with a fixed, orthonormal discrete wavelet basis. Each block first maps the input window through a RootBlock variant, producing a hidden representation $\mathbf{h}$. Separate trainable linear heads then produce backcast and forecast coefficient vectors,

$$\boldsymbol{\theta}^b = \mathbf{W}_b \mathbf{h}, \qquad \boldsymbol{\theta}^f = \mathbf{W}_f \mathbf{h}$$

For each target length $T \in \{L, H\}$, a wavelet synthesis matrix is constructed at model initialization by applying inverse DWT reconstruction to unit impulses in each wavelet coefficient band. This raw synthesis matrix is then orthogonalized via SVD, and the rows of $V^\top$ form the fixed orthonormal bases $\boldsymbol{\Psi}_b \in \mathbb{R}^{k \times L}$ and $\boldsymbol{\Psi}_f \in \mathbb{R}^{k \times H}$ (one per path). The basis expansion generators therefore evaluate to

$$g_b(\boldsymbol{\theta}^b) = \boldsymbol{\theta}^b \, \boldsymbol{\Psi}_b = \hat{\mathbf{x}}, \qquad g_f(\boldsymbol{\theta}^f) = \boldsymbol{\theta}^f \, \boldsymbol{\Psi}_f = \hat{\mathbf{y}}$$

Because $\boldsymbol{\Psi}_b$ and $\boldsymbol{\Psi}_f$ are frozen at initialization and stored as non-trainable buffers, inference requires only a single matrix multiply per path, which GPUs excel at. A full DWT is never computed at training or inference time. Implementation details are given in Appendix B.

2.3.1 Composition blocks: TrendWavelet and TrendWaveletGeneric

The strongest single-block family in our benchmarks combines a frozen polynomial trend basis with a frozen orthonormal wavelet basis in a single additive decomposition. The `TrendWavelet` block produces the coefficient vector $\boldsymbol{\theta} \in \mathbb{R}^{p+k}$ from the shared hidden representation $\mathbf{h}$, then partitions and expands it:

$$\hat{\mathbf{y}} = \underbrace{\boldsymbol{\theta}^f_{1:p} \, \mathbf{T}_f}_{\text{trend}} + \underbrace{\boldsymbol{\theta}^f_{p+1:p+k} \, \boldsymbol{\Psi}_f}_{\text{wavelet}}$$

where $\mathbf{T}_f \in \mathbb{R}^{p \times H}$ and $\mathbf{T}_b \in \mathbb{R}^{p \times L}$ are the Vandermonde polynomial bases of order p (`trend_thetas_dim`), $\boldsymbol{\Psi}_f \in \mathbb{R}^{k \times H}$ and $\boldsymbol{\Psi}_b \in \mathbb{R}^{k \times L}$ are the orthonormal wavelet bases of rank k (`basis_dim`), and the same additive structure holds for the backcast path. The two bases are complementary by construction: the polynomial captures smooth low-frequency trend while the wavelet captures localized oscillatory residuals.

`TrendWaveletGeneric` extends this to a three-way decomposition by appending a third learned (data-driven) branch of rank r (`generic_dim`):

$$\hat{\mathbf{y}} = \boldsymbol{\theta}^f_{1:p} \, \mathbf{T}_f + \boldsymbol{\theta}^f_{p+1:p+k} \, \boldsymbol{\Psi}_f + \boldsymbol{\theta}^f_{p+k+1:p+k+r} \, \mathbf{G}_f$$

where $\mathbf{G}_f \in \mathbb{R}^{r \times H}$ is a trainable rank-r basis matrix. The two structured bases remain frozen; only coefficients and the generic basis are learned. This gives the block a fall-back data-driven capacity for signal components that neither polynomial nor wavelet bases can represent compactly.

Both block families are fully composable with both AE backbone variants (AERootBlock, AERootBlockLG), yielding the `TrendWaveletAE`, `TrendWaveletAELG`, `TrendWaveletGenericAE`, and `TrendWaveletGenericAELG` configurations benchmarked in Section 4.

2.3.2 Asymmetric and tiered wavelet bases

Two additional hyperparameters control how the wavelet basis is oriented relative to the backcast and forecast windows.

Asymmetric wavelet families. When backcast and forecast lengths differ substantially — as in Traffic-96 ($L = 480$, $H = 96$, ratio $5\times$) — the optimal wavelet support length differs between paths. The `backcast_wavelet_type` and `forecast_wavelet_type` parameters allow independent wavelet family selection per path. A long-support wavelet (e.g. Symlet-20) captures broad low-frequency structure in the long backcast; a short-support wavelet (e.g. Daubechies-3, Coiflet-2) provides compact localized bases for the short forecast. When these overrides are omitted, both paths share the same `wavelet_type`.

Tiered basis offset. The `basis_offset` parameter shifts the starting row of the basis matrix selected from $V^\top$, cycling through the SVD spectrum in successive stacks. In a depth-S model, stack ℓ uses rows $[\ell \cdot \Delta, \ell \cdot \Delta + k)$ of the orthonormal basis, where Δ is a stride derived from `basis_offset`. This spreads each stack's basis coverage across different frequency bands, encouraging the doubly residual stack to decompose the signal rather than repeatedly fitting the same spectral region. Tiered offsets are most effective on the Daily M4 period, where they appear in 8 of the top-10 configurations (Appendix D).

3 Experimental Setup

3.1 Datasets

M4 [Makridakis et al., 2020] is the primary benchmark (six frequencies, 100,000 series, $H \in \{6, 8, 24, 13, 14, 48\}$). We complement M4 with four out-of-distribution datasets: Tourism (1,311 series, Y/Q/M), PeMS Traffic (862 hourly sensors), Weather (21 channels at 10-min resolution), and the univariate Milk production series.

Table 4: Datasets and evaluation parameters. The paper-sample protocol enforces the in-sample constraint $L_h \cdot H$ [Oreshkin et al., 2020, Appendix D] on the trailing window of each series, with $L_h = 1.5$ for Yearly/Quarterly and $L_h = 10$ otherwise.

Dataset	Frequency	Series	Horizon H	Lookback L	Primary metric
M4 (Y/Q/M/W/D/H)	mixed	100,000	$\{6, 8, 24, 13, 14, 48\}$	$L = 5H$	sMAPE, OWA
Tourism (Y/Q/M)	mixed	1,311	$\{4, 8, 24\}$	$L = 5H$	sMAPE
Traffic-96	hourly	862	96	$5H$	MSE, MAE
Weather-96	10-min	21 ch.	96	$5H$	MSE, MAE
Milk	monthly	1	6	$5H$	sMAPE

3.2 Training Protocols

Every architecture is trained under three complementary protocols (sliding-window with cosine LR; the iteration-based paper-sampling protocol of Oreshkin et al. [2020] with MultiStepLR; and the same paper-sampling protocol with `ReduceLROnPlateau` swapped in) over 10 random seeds per (configuration, period, protocol) cell, with pairwise comparisons by Wilcoxon signed-rank test under Bonferroni correction. Full settings and verbatim YAML are in Appendix A.

Table 5: Protocol-local winners used for the main comparison. M4 rows use the paper-sample protocol; Weather-96 uses the comprehensive-sweep protocol. Tourism and Milk are summarized in Appendix F. Scores are mean $\pm$ std over the surviving seeds. Asterisks mark `active_g=forecast`, which is not part of the original N-BEATS paper [Oreshkin et al., 2020].

Task	Metric	Best config	Score	Params	n	N-BEATS ref.
M4-Yearly	sMAPE	`T+Coif2V3 30s bdeq`	13.542 ± 0.148	15.2M	10	G-10: 13.564
M4-Quarterly	sMAPE	`NBEATS-IG 10s ag0`	10.312 ± 0.055	19.6M	10	same
M4-Monthly	sMAPE	`TW 30s td3 sym10`	13.240 ± 0.334	6.8M	9	IG-30: 13.307
M4-Weekly	sMAPE	`T+Coif2V3 30s bdeq`	6.735 ± 0.203	15.8M	10	IG-10: 6.974
M4-Daily	sMAPE	`T+Sym10V3 10s tiered`	3.014 ± 0.035	5.3M	10	IG-30: 3.062
M4-Hourly	sMAPE	`NBEATS-IG 30s agf`*	8.758 ± 0.099	43.6M	10	IG-30: 8.906
Weather-96	MSE	`TAE+DB3V3AE 30s ld8`	0.138 ± 0.027	7.1M	10	IG-10: 0.183

3.3 Architecture Sweep and Cross-Architecture Transfer

The central hypothesis is that orthonormal wavelet bases and autoencoder bottlenecks enable 60–95% parameter compression relative to paper-faithful Generic stacks while matching or exceeding their forecasting accuracy. We evaluate this under a single 112-configuration *Comprehensive Sweep* (sliding-window protocol) plus a curated 53-configuration subset re-run under both paper-sample variants. The five sweep groups (paper baselines, homogeneous `TrendWavelet` stacks, alternating `T+W` dual-stack patterns, efficiency bottlenecks, and Generic-AE controls) are enumerated in Appendix C; verbatim YAMLs and run scripts ship with the submission. We additionally sweep `active_g` $\in \{\text{False}, \text{forecast}\}$ on the Generic and GenericAELG families to isolate that `lightningnbeats`-extension knob from the wavelet and AE structure.

To confirm the proposed blocks are not specific to the N-BEATS doubly-residual stack, we re-run curated subsets on two other backbones. NHiTS [Challu et al., 2023] adds multi-rate input pooling and hierarchical forecast interpolation around the same block interface; we sweep on Weather-96 and Traffic at $H \in \{96, 192, 336, 720\}$ (MSE loss, 100 epochs, 8 seeds). For Transformer transfer (Section 4.5) we replace each Llama attention projection with a frozen `TrendWavelet` basis on the input-feature axis and the SwiGLU MLP with the same hourglass autoencoder used in our `AERootBlock`, then pretrain a SmolLM2-135M-class model [Allal et al., 2025] from scratch on FineWeb-Edu [Penedo et al., 2024] under a 3.0 B-token budget per variant. When block-level effects survive the backbone change, the structured priors are driving them rather than the residual scheduling.

4 Results

All tables apply the strict health filter used in the analysis reports: a run is excluded if `diverged=True`, if sMAPE exceeds 100, or if `best_epoch=0` with sMAPE above 50. Means and standard deviations are computed over surviving seeds. For M4, the main comparison uses the paper-sample protocol to stay close to the N-BEATS training setup of Oreshkin et al. [2020]; sliding-window M4 results, plateau-LR variants, and full per-period tables are deferred to Appendix E because absolute sMAPE is not comparable across sampling protocols.

4.1 Per-Period Leaderboards

Table 5 gives the compact leaderboard used for the main claims. Wavelet and TrendWavelet variants win or tie most short- and medium-horizon tasks, but the result is not universal: N-BEATS-I+G [Oreshkin et al., 2020] remains the M4-Quarterly and M4-Hourly reference. M4-Hourly also illustrates an important convention: the winning `agf` configuration uses `active_g=forecast`, a `lightningnbeats` extension rather than the paper-faithful N-BEATS block.

The strongest cross-dataset statement is therefore conditional rather than absolute: structured wavelet bases are a better use of parameters on most tasks in the sweep, but the winning architecture changes with horizon, sampling protocol, and dataset size. Tourism and Milk show the same pattern at smaller scale and are reported in Appendix F.

Table 6: Best sub-1M M4 configurations under the paper-sample protocol. "Gap" is relative to the period winner in Table 5; "Param ratio" compares the period winner's parameter count with the compact model.

Period	Compact config	Params	sMAPE	Gap	Param ratio
Yearly	`TWAE 10s ld32 ag0`	0.48M	13.546 ± 0.102	+0.03%	32×
Quarterly	`TWAE 10s ld32 ag0`	0.49M	10.404 ± 0.028	+0.89%	40×
Monthly	`TWAE 10s ld32 sym10`	0.58M	13.513 ± 0.378	+2.06%	12×
Weekly	`TWAELG 10s ld32 db3`	0.54M	7.252 ± 0.263	+7.68%	29×
Daily	`TWGAELG 10s ld16 db3`	0.52M	3.051 ± 0.049	+1.21%	10×
Hourly	`TWAELG 10s ld32 db3 agf`*	0.85M	8.924 ± 0.129	+1.89%	51×

4.2 Parameter-Efficiency Frontier

Parameter efficiency is the most stable positive result and is the direct test of the central hypothesis stated in Section 3. Table 6 isolates the best sub-1M M4 model for each paper-sample period. These models are not always winners — especially on Weekly — but they repeatedly sit close to the top of the leaderboard at one-tenth to one-fiftieth the parameter count. The clearest example is M4-Hourly: `TWAELG_10s_ld32_db3_agf` uses 0.85M parameters and trails the 43.6M `NBEATS-IG_30s_agf` reference by only 0.166 sMAPE.

4.3 Stability and Ablations

The negative results are as important as the winners. Overparameterized Generic stacks are unstable: `NBEATS-G_30s_ag0` (paper baseline group of Section 3) has bimodal paper-sample behavior on M4-Quarterly (15.40 ± 9.97), M4-Weekly (8.78 ± 4.61), and M4-Daily (10.21 ± 12.01), even though its best seeds can look competitive. TrendWavelet blocks do not show the same stuck-mode behavior. The small-data version of the same effect shows up on Milk (Appendix F), where AE-family TrendWavelet models cut divergence to the low single digits.

The ablations support five practical rules; the underlying matched-seed Wilcoxon tables are listed in Appendix D.

1. `active_g=forecast` is not a global default. It is decisive on M4-Hourly (9/9 matched pairs improve, 5 significant at $p < 0.05$) and rescues some Generic convergence failures, but is worse or neutral elsewhere and harmful for Weather unified stacks.
2. AE and AELG are statistically similar on M4 at matched configurations; AELG is preferred only when its smaller latent dimension is needed for parameter budget.
3. Tiered basis offsets are strongest on paper-sample M4-Daily but regress on Weekly, so they belong to a Daily-specific recipe rather than the main default.
4. Wavelet family matters across datasets, but the M4 shortlist reduces to `haar`, `db3`, `coif2`, and `sym10`; `coif3` has no per-period M4 SOTA and is dropped from the main defaults.
5. Backbone preference reverses by data regime: RootBlock variants are strongest on M4, while AE variants lead on Weather. Tourism and Milk reproduce this pattern at smaller scale (Appendix F).

4.4 Cross-Architecture Transferability: NHiTS

The block interface transfers more cleanly than the hyperparameters. On the NHiTS backbone [Challu et al., 2023] TrendWavelet and BottleneckGenericAE/AELG families dominate the short-epoch Weather sweep when substituted for the vanilla NHiTS Generic block, with the pooling and hierarchical interpolation schedules unchanged. Short-horizon NHiTS ($H = 96$) favors unified `TrendWavelet`, while long horizons ($H = 336, 720$) favor `BottleneckGenericAE`; `active_g=forecast`, decisive on M4-Hourly, is neutral-to-negative here, and `TrendWaveletGenericVAE` is unexpectedly competitive but we treat that as NHiTS-specific. The cross-architecture takeaway is narrow but genuine: the structured wavelet and AE/AELG components remain useful when lifted out of the N-BEATS doubly-residual schedule, while optimal stack pattern, latent dimension, and scheduler stay dataset- and architecture-dependent. Matching Traffic horizons under the same harness are in Appendix D.

4.5 Cross-Architecture Transferability: Transformer LLMs

The same blocks port into a Transformer with no change in shape. We replace each Llama attention projection with a frozen `TrendWavelet` basis on the input-feature axis, and the SwiGLU MLP with the same hourglass autoencoder used in our `AERootBlock`. Pretraining a SmolLM2-135M-class model [Allal et al., 2025] from scratch on FineWeb-Edu [Penedo et al., 2024] under a 3.0 B-token budget, the wavelet variant `tw_root_fc_db3_64_tiered_silu` (67.2M params, 50.0% compression) reaches a calibrated 24.4 val PPL, Pareto-dominating the pure AE-MLP baseline (69.3M, 26.45) against the 134.5M / 19.82 vanilla reference. A frozen basis on its own is not enough: pure `TrendWavelet` on attention is Pareto-dominated by AE-MLP despite retaining 17% more parameters, and a pure-basis MLP with no learned reduction collapses outright at 250 M tokens. Adding a learned $\mathrm{SiLU}(\mathbf{W}\mathbf{x})$ stage in front of the basis restores competitiveness, the same lesson Section 4.3 draws on M4: a capacity reduction needs a structural prior the optimizer can actually reach. With a pretrained teacher available, the same AE-LG bottleneck at roughly 90% MLP-block compression on Llama-3.2-1B reaches 18.81 PPL after AE pretraining and KD fine-tuning, beating the unmodified 19.45 PPL original. As on M4, where the savings make sub-1M-parameter winners like `TWAELG_10s_ld32_db3` possible (Table 6), in the Transformer the freed budget can be redeposited as deeper stacks, richer composite bases, or wider wavelet sets; full architecture port, trajectories, and distillation pipeline are in Appendix G.

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A Paper-Sample Protocol Details

This appendix documents the two paper-sample training protocols in full so that the M4 numbers in Section 4 can be reproduced from a single YAML file. Both variants share the sampling strategy and step budget of the original N-BEATS training procedure; they differ only in the learning-rate scheduler.

A.1 Faithful replication (MultiStepLR)

The faithful variant (`comprehensive_m4_paper_sample.yaml`) is intended as a direct copy of the training procedure described in Oreshkin et al. [2020, Section 5.2 and Appendix D], with no deviations except the small validation-cadence correction described below.

Sampling. Each gradient step draws a batch of 1024 (series, anchor) pairs uniformly at random *with replacement* from the training pool. A series is eligible at step t if it has at least $L + H$ observations; for a chosen series of length T, the anchor index is sampled uniformly from $[T - L_h \cdot H,\ T - H]$, restricting valid windows to the trailing $L_h \cdot H$ time steps. Per-period L_h is fixed to the values in Table 4: $L_h = 1.5$ for Yearly and Quarterly, $L_h = 10$ for Monthly, Weekly, Daily, and Hourly. The lookback length is $L = 5H$.

Optimization. Adam with $\eta_0 = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, no weight decay. sMAPE loss is computed per-step on the 1024-window batch. The gradient budget is 10^5 total steps, with one Lightning epoch defined as 1,000 steps (≈100 paper-epochs).

Learning-rate schedule. MultiStepLR with 10 evenly spaced milestones at $\{10\text{k}, 20\text{k}, \ldots, 100\text{k}\}$ steps, $\gamma = 0.5$. The LR is therefore halved at each 10% boundary of the training budget. This is exactly the schedule used by Oreshkin et al.

Validation and early stopping. Validation is run every 100 training steps (`val_check_interval=100`) on the held-out window of each series. Early stopping monitors validation sMAPE with patience 15 (counted in val-check units, so a minimum of 1,500 steps must elapse before stopping can fire) and $\Delta_{\min} = 10^{-3}$. The sub-epoch validation cadence is the only departure from a strict line-by-line copy of Oreshkin et al. [2020]; without it, early stopping fires on the first warmup-corrupted check and collapses to `best_epoch` $\in \{0, 1\}$.

Other settings. `n_blocks_per_stack = 1`, `share_weights = true`, ReLU activations throughout, `forecast_multiplier = 5`, default block widths (`g_width = 512`, `s_width = 2048`, `t_width = 256`).

A.2 Plateau LR variant (ReduceLROnPlateau)

The plateau variant (`comprehensive_m4_paper_sample_plateau.yaml`) is identical to A.1 in every respect except the learning-rate schedule. The motivation is that MultiStepLR commits to a fixed decay schedule independent of how training is actually progressing. A plateau scheduler can react to validation stalls and find a more aggressive effective decay on cells where the loss surface is well-behaved.

Learning-rate schedule. `ReduceLROnPlateau` with `factor = 0.5`, `patience = 3` (val-check units, so the LR is reduced after 3 consecutive 100-step val checks without improvement of at least $\Delta_{\min} = 10^{-3}$), `cooldown = 1` (one val-check skipped after each reduction before monitoring resumes), `min_lr =` 10^{-6}, `mode = min`, monitored on `val_loss`. The scheduler is stepped on the same cadence as validation (every 100 training steps) so that its trigger logic operates in the same time units as early stopping.

Effect. Empirically the plateau scheduler delivers more LR reductions than MultiStepLR on M4 Quarterly, Monthly, and Daily, and fewer on Yearly and Weekly, which matches the observed differences in the per-period leaderboards (Section 4).

All other settings (sampling, L_h, batch size, step budget, sub-epoch validation, early stopping, optimizer, loss, block widths, weight sharing, seed budget) are inherited unchanged from A.1.

A.3 Why two paper-sample protocols

The N-BEATS paper reports a single training procedure, so any one-protocol comparison conflates the architectural change with whatever artifacts that specific schedule introduces on the new architecture. By running each architecture under both the faithful protocol (Appendix A, MultiStepLR) and the plateau variant (Appendix A, ReduceLROnPlateau) we can attribute observed gains to one of three categories. (i) Gains under both protocols are robust evidence for the architectural claim. (ii) Gains under the faithful protocol only constitute a head-to-head replacement for the published baseline. (iii) Gains under the plateau protocol only are a finding contingent on a more responsive LR schedule, which we report as such rather than as a clean architectural win.

B Hyperparameters and Empirical Defaults

This appendix is the single reference for every hyperparameter exposed by the `lightningnbeats` model classes used in this paper. For each hyperparameter we report (i) the constructor default in the released package, (ii) the value used in the M4 sweeps reported in Section 4, (iii) the swept range when applicable, and (iv) the empirical recommendation that came out of those sweeps. Recommendations are M4-only; out-of-distribution dataset deltas (Tourism, Traffic, Weather, Milk) are summarized in Section 3 and not repeated here. Reproducibility pointers to canonical CSVs and YAML configs are collected in Section B.9.

B.1 Code defaults vs paper-sweep values

Table 7 enumerates every keyword argument of `NBeatsNet.__init__` (`src/lightningnbeats/models.py`, lines 88–313). The constructor defaults define the released package surface; the paper-sweep column lists the value or grid actually used in the experiments of Section 3. Where the paper-sweep value disagrees with the constructor default — most notably `share_weights` — the override is documented at Appendix A (see "Other settings").

Lookback and forecast multiplier. For all M4 cells we use $L = 5H$ (`forecast_multiplier=5`). The paper-sample protocol additionally clamps the valid sampling window to the trailing $L_h \cdot H$ time steps with $L_h = 1.5$ on Yearly/Quarterly and $L_h = 10$ elsewhere; see Appendix A.

B.2 Architectural sweep grids

Table 8 lays out the grid actually executed for each block family in the 112-config Comprehensive Sweep (sliding) and the curated 53-config subset (paper-sample MultiStepLR + plateau). Block-family abbreviations match Section 3: `T+W` = alternating `Trend` + `WaveletV3` (RootBlock), `T+W AELG` = alternating `TrendAELG` + `WaveletV3AELG`, `TW` = unified `TrendWavelet` block, `TWAE/TWAELG` = unified `TrendWaveletAE/TrendWaveletAELG`, `TWGAELG` = `TrendWaveletGenericAELG`.

Wavelet basis implementation. For each target length $T \in \{L, H\}$, `_WaveletGeneratorV3` (`src/lightningnbeats/blocks/blocks.py:2621`) builds a raw synthesis matrix of size $T \times T$ by reconstructing each unit impulse in each DWT coefficient band through `pywt.waverec`. The

Table 7: N-BEATS / lightningnbeats hyperparameters: constructor defaults and the values used in this paper's M4 sweeps.

Parameter	Constructor default	Paper sweep value (M4)
Optimization (`_NBeatsBase`)		
`loss`	`SMAPELoss`	`SMAPELoss`
`optimizer_name`	`Adam`	`Adam` (β_1=0.9, β_2=0.999)
`learning_rate`	10^{-3}	10^{-3}
`sum_losses`	`False`	`False`
`kl_weight`	10^{-3}	0.1 for VAE families only
Stack composition (`NBeatsNet`)		
`stack_types`	*required*	per-config (Section 3)
`n_blocks_per_stack`	1	1
`share_weights`	`False`	`True` (paper-sample, Appendix A)
`active_g`	`False`	swept over {`False`, `'forecast'`}
Block widths		
`g_width`	512	512
`s_width`	2048	2048
`t_width`	256	256
`ae_width`	512	512
Block-level priors		
`latent_dim`	5	swept $\{8, 16, 32\}$
`basis_dim`	32	set per-period via `eq_fcast` or $2 \cdot$ `eq_fcast`
`forecast_basis_dim`	`None`	not used on M4
`trend_thetas_dim`	3	swept $\{3, 5\}$
`wavelet_type`	`db3`	swept {haar, db3, coif2, sym10}
`backcast_wavelet_type`	`None`	not used on M4 (Traffic-96 only)
`forecast_wavelet_type`	`None`	not used on M4 (Traffic-96 only)
Cross-stack residuals		
`skip_distance`	0	0
`skip_alpha`	0.1	not exercised on M4

Table 8: Hyperparameter grids per block family. "–" means the parameter does not apply to the family.

Family	n_{stacks}	wavelet	`trend_thetas_dim`	`basis_dim`	`latent_dim`	`active_g`	`share_weights`
`NBEATS-G`	$\{10, 30\}$	–	–	–	–	{F, f}	`True`
`NBEATS-IG`	$\{10, 30\}$	–	3	–	–	{F, f}	`True`
`TW`	$\{10, 30\}$	shortlist	$\{3, 5\}$	{eq, 2·eq}	–	`F`	`True`
`T+W` (RB)	$\{10, 30\}$	shortlist	3	{eq, 2·eq}	–	{F, f}	`True`
`T+W AELG`	$\{10, 30\}$	shortlist	3	`eq`	$\{16, 32\}$	{F, f}	`True`
`TWAE`	10	shortlist	3	`eq`	$\{8, 16, 32\}$	{F, f}	`True`
`TWAELG`	10	shortlist	3	`eq`	$\{16, 32\}$	{F, f}	`True`
`TWGAELG`	10	shortlist	3	`eq`	$\{16, 32\}$	{F, f}	`True`

Wavelet shortlist: {haar, db3, coif2, sym10}. *eq* $\equiv$ *forecast_length.* F $\equiv$ `False`, f $\equiv$ `'forecast'`.

decomposition level is $\min(L_{\max}, 5)$, where $L_{\max} =$ `pywt.dwt_max_level`$(T,$ `dec_len`$)$; when $L_{\max} = 0$ (filter longer than target), we keep level 0 rather than forcing a boundary-corrupted level-1 decomposition. The raw matrix is then SVD-orthogonalized: rows of $V^\top$ form an orthonormal basis ordered low-to-high frequency by singular value. `basis_offset` chooses the starting row and `basis_dim` the row count, so a stack at offset $\ell\,\Delta$ with width k uses $V^\top[\ell\,\Delta : \ell\,\Delta + k,\ :]$. The result is registered as a non-trainable buffer; inference is a single `matmul`.

B.3 Training-protocol hyperparameters (sliding)

The faithful and plateau paper-sample variants are documented in Appendix A. The sliding-window protocol used by the Comprehensive Sweep (`experiments/configs/comprehensive_sweep_m4.yaml`) is: epoch-based training with

sMAPE loss, Adam at 10^{-3}, batches drawn by sliding the lookback over each series; `max_epochs`$=$ 75, `min_epochs`$=$ 10, early-stop `patience`$=$ 20 with $\Delta_{\min} = 10^{-3}$; cosine annealing with a 15-epoch linear warmup down to $\eta_{\min} = 10^{-6}$. Sub-epoch validation is not required under sliding because the per-epoch validation pass is on the natural held-out tail of each series.

B.4 Per-period recommendations

Sections B.4.1–B.4.3 are the empirical findings that came out of the sweeps in Section 4. Numbers are taken verbatim from the meta-analysis report `experiments/analysis/analysis_reports/m4_overall_leaderboard_2026-05-03.md`; absolute sMAPE is not comparable across protocols.

B.4.1 LR scheduler by period and protocol

Table 9: Recommended LR scheduler per (period, protocol). Plateau $\equiv$ `ReduceLROnPlateau` with patience $= 3$ val-check units, factor $= 0.5$, cooldown $= 1$, min_lr $= 10^{-6}$. Step-paper $\equiv$ MultiStepLR with 10 evenly spaced milestones, $\gamma = 0.5$. Cosine-warmup $\equiv$ 15-epoch linear warmup then cosine annealing to $\eta_{\min} = 10^{-6}$.

Period	Paper-sample (`nbeats_paper`)	Sliding
Yearly	step-paper (neutral; either works)	cosine-warmup
Quarterly	**plateau** ($\Delta_{\text{step}\to\text{plateau}} = -0.029$, $n = 30$)	cosine-warmup
Monthly	**plateau** ($\Delta = -0.089$, median -0.198, $n = 18$)	cosine-warmup
Weekly	step-paper (plateau regresses by $+0.20$ at $n = 10$, locked out)	cosine-warmup
Daily	**plateau** ($\Delta = -0.028$, $n = 13$)	cosine-warmup
Hourly	step-paper (head-to-head against plateau not yet run on novel families)	cosine-warmup

Mandatory. Plateau LR on `nbeats_paper` requires sub-epoch validation: `val_check_interval=100`, `patience=20` val-check units, $\Delta_{\min} = 10^{-3}$. Without these, early stopping fires on the first warmup-corrupted check and collapses to `best_epoch` $\in \{0, 1\}$ (Appendix A).

B.4.2 Sampling protocol verdict per period

Table 10: Sampling-protocol verdict per M4 period. $\Delta =$ sliding $-$ paper-sample on the best config for each protocol; negative Δ favors sliding. Absolute sMAPE within a row is not comparable across protocols (Appendix A).

Period	H	Best sliding sMAPE	Best paper-sample sMAPE	Δ	Winner
Yearly	6	13.499	13.542	≈ 0	tie
Quarterly	8	10.127	10.313	-0.19	**sliding**
Monthly	18	13.279	13.240	$+0.04$	paper-sample
Weekly	13	6.671	6.735	-0.06	sliding
Daily	14	2.588	3.012	$\mathbf{-0.42}$	**sliding**
Hourly	48	8.587	8.758	-0.17	**sliding**

For paper-faithful comparison to Oreshkin et al. [2020] we report paper-sample numbers in the main body. For absolute-sMAPE leaderboards on long-horizon periods (Daily, Hourly, Quarterly), sliding is the stronger protocol. We never mix the two within a leaderboard.

B.4.3 Wavelet family per period

The shortlist {haar, db3, coif2, sym10} is the M4 default. `coif3` is dropped because it produced no per-period SOTA in either protocol despite being tested. Per-period leaders are summarized in Table 11. The cross-period generalist under paper-sample is `sym10` (configuration `T+Sym10V3_10s_tiered_ag0`, mean rank $13.33/108$); under sliding it is `haar` (`T+HaarV3_30s_bd2eq`, mean rank $12.7/112$).

Table 11: Best wavelet family per M4 period within the shortlist. Paper-sample uses the LR scheduler from Table 9; sliding uses cosine-warmup throughout.

Period	Paper-sample best wavelet	Sliding best wavelet
Yearly	db3 / coif2 (within seed noise)	coif2
Quarterly	sym10 (paper baseline NBEATS-IG wins overall)	sym10 / haar
Monthly	sym10 (unified TW)	coif2 (unified TW)
Weekly	coif2 (alternating T+W)	db3 (alternating T+W)
Daily	sym10 (tiered)	paper baseline (NBEATS-G)
Hourly	sym10 (tiered) or paper baseline	paper baseline (NBEATS-IG)

B.5 Block-level recommendations

The following compact rules summarize the empirical findings for each block-level hyperparameter on M4. Each rule is sourced from the meta-analysis report and the sweep CSVs in `experiments/results/m4/`.

`basis_dim`**.** Use the label `eq_fcast`, i.e. $k = H$, as the default. At convergence (R3, ≥ 50 epochs equivalent) the spread between `eq_fcast`, `lt_bcast`, and `eq_bcast` is < 0.014 sMAPE on M4-Yearly across 14 wavelet families and three datasets; the apparent advantage of larger labels at 10 epochs is a convergence-speed artifact. Avoid `lt_fcast` for $H \leq 8$ (0% R3 survival on M4-Yearly in the AELG sweep). For unified `TrendWavelet` blocks the basis-dim sensitivity is higher (spread 0.37 sMAPE), so `eq_fcast` is even more critical there.

`latent_dim`**.** For `AERootBlockLG`-backed blocks (`TWAELG`, `TAELG+WV3AELG`, etc.), `latent_dim`$= 16$ is the default ($n \geq 600$ runs, $p < 0.02$ vs 32 on both M4-Yearly and Tourism-Yearly). For plain `AERootBlock`, `latent_dim`$\in \{5, 8\}$ are statistically equivalent (Mann–Whitney $p = 0.71$, $n = 330$ each); `latent_dim`$= 2$ is catastrophically underparameterized ($p < 3 \times 10^{-20}$ vs $\{5, 8\}$). The unified `TrendWaveletAELG` block prefers `latent_dim`$= 16$, with one known catastrophic combination: `db4` + `eq_fcast` + `latent_dim`$= 16$ (sMAPE ~ 76); `db4` at `latent_dim`$= 8$ or with other basis-dim labels is fine.

`trend_thetas_dim`**.** Use 3 for $H \leq 10$ and 5 for $H \geq 50$. Significant on Tourism-Yearly ($H = 4$, $p = 0.008$, ttd$= 3$) and on Weather-96 ($H = 96$, $p = 0.042$, ttd$= 5$) at convergence. Mid-horizon crossover (Monthly $H = 18$, Weekly $H = 13$, Daily $H = 14$) defaults to 3 on insufficient evidence. The earlier "always ttd$= 3$" finding was a 10-epoch convergence-speed artifact on Weather, reversed at 50 epochs.

Trend backbone hierarchy. On M4 the ordering is RootBlock $>$ AELG $>$ AE for alternating Trend+Wavelet stacks. The plain AE bottleneck homogenizes wavelet-family signal in alternating configurations ($\eta^2 = 0.003$, ns) and is not viable; the learned gate in AELG restores competitiveness. On Weather-96 and Milk this hierarchy reverses (AE $>$ AELG, Mann–Whitney $p = 0.036$ on Weather), but those datasets are out of scope for this appendix.

Block ordering. Always place `Trend*` stacks before `Wavelet*`, `Generic*`, or `Seasonality*` stacks. In the VAE2 study (M4-Yearly + Weather-96) every significantly-different ordering pair — including one Bonferroni-significant pair on M4 (Cohen's $r = 0.83$) — favored Trend-first; no pair favored the reverse. For VAE-family backbones, reversing the order can $2{-}3\times$ the OWA. Treat this as a fixed convention, not a hyperparameter to search.

TrendWavelet vs alternating T+W. Under paper-sample protocol on M4, alternating $T{+}\langle\text{wav}\rangle$V3 (RootBlock) beats unified `TrendWavelet` by ≈ 10 mean-rank points (14.1 vs 24.3 across 53 configs $\times 6$ periods). Default to alternating when ≥ 15M parameters is acceptable. Reserve unified `TrendWavelet` (or its AE/AELG variants) for parameter-constrained scenarios: `TWAELG_10s_ld32_db3` achieves 0.48–0.85M parameters and top-5 on Yearly, Daily, and Hourly. Plateau LR has flipped the Monthly winner to unified `TW_30s_td3_bdeq_sym10` (13.240); on every other period alternating is at least co-optimal.

`active_g`. The constructor default `False` (paper-faithful, abbreviated `ag0`) is the robust choice on M4. `active_g='forecast'` (`agf`) helps on Hourly only — every paper backbone gains 0.07–0.15 sMAPE there — and ties or loses on every other period. `active_g` is a `lightningnbeats` extension and is not part of the original N-BEATS architecture: configurations marked `ag0` are paper-faithful, configurations marked `agf` apply the extension and are not strictly comparable to Oreshkin et al. [2020].

Other ablations (not exercised on M4). `skip_distance` (cross-stack residual injection) never helps on M4 in the configurations tested and slightly hurts deep stacks; `sum_losses` adds a backcast-reconstruction term ($0.25 \cdot \mathrm{sMAPE}(\hat{x}, 0)$) and is reserved for the convergence study reported in Appendix D. Both default to off in this paper.

B.6 Drop list

Configurations excluded *a priori* from the curated paper-sample subset and from any paper-headline leaderboard. Each is supported by at least one CSV in `experiments/results/m4/`.

- **BottleneckGeneric family** (`BNG*`, `BNAE*`, `BNAELG*`): universally worst on M4.
- **Pure VAE configs** (`GenericVAE_3s_sw0`, etc.): mean sMAPE 55–68 on Q/M/W/D, unusable.
- `NBEATS-G_30s_ag0` **on Q/W/M**: bimodal collapse, sMAPE std 7.4–9.5.
- **All** `*_sd5` **variants** (`skip_distance`$= 5$): never help on M4.
- **All** `*_coif3` **variants on M4**: no per-period SOTA across either protocol.
- `_30s_agf` **tiered configs at** `_10s` **depth**: run-to-run divergence outliers.
- **Step-paper LR on Q/M/D**: dominated by plateau LR (Table 9).
- **Pure** `GenericAE_*`**,** `GenericAELG_*`: bottoms out at mean rank ≈ 38.

B.7 Quick-pick decision table

Table 12 is the per-period recipe to copy into a YAML config for a new M4 experiment. Each row reproduces the per-period top cell of the appropriate leaderboard (paper-sample if the column says `nbeats_paper`, sliding otherwise) from `m4_overall_leaderboard_2026-05-03.md`.

Table 12: Best M4 configuration per (period, protocol). All cells are $n = 10$ unless noted. "ttd" = `trend_thetas_dim`; "ld" = `latent_dim`; "bd" = `basis_dim` label; "ag" = `active_g` (0 = `False`, f = `'forecast'`).

Period	Protocol	LR	Block family / config	Depth	Wavelet	ttd	bd	ld	ag	sMAPE
Yearly	paper-sample	plateau	`T+Coif2V3` (RB)	30	coif2	3	`eq`	–	0	13.542
Quarterly	paper-sample	plateau	`NBEATS-IG`	10	–	3	–	–	0	10.313
Monthly	paper-sample	plateau	`TW` (unified)	30	sym10	3	`eq`	–	0	13.240
Weekly	paper-sample	step-paper	`T+Coif2V3` (RB)	30	coif2	3	`eq`	–	0	6.735
Daily	paper-sample	plateau	`T+Sym10V3` (RB, tiered)	10	sym10	3	`eq`	–	0	3.012
Hourly	paper-sample	step-paper	`NBEATS-IG`	30	–	3	–	–	f	8.758
Yearly	sliding	cos-warmup	`TW` (unified)	10	coif2	3	`eq`	–	0	13.499
Quarterly	sliding	cos-warmup	`NBEATS-IG`	10	–	3	–	–	0	10.127
Monthly	sliding	cos-warmup	`TW` (unified)	30	coif2	3	`2eq`	–	0	13.279
Weekly	sliding	cos-warmup	`T+Db3V3` (RB)	30	db3	3	`eq`	–	0	6.671
Daily	sliding	cos-warmup	`NBEATS-G`	30	–	–	–	–	0	2.588
Hourly	sliding	cos-warmup	`NBEATS-IG`	30	–	3	–	–	f	8.587
Sub-1M parameter champions (paper-sample):										
Yearly	paper-sample	plateau	`TWAE_10s_ld32`	10	db3	3	`eq`	32	0	13.546
Daily	paper-sample	plateau	`TWGAELG_10s_ld16_db3`	10	db3	3	`eq`	16	0	3.051

B.8 Cross-period generalists

When a single configuration must run across all six M4 periods (cross-period reporting, ensemble bases, ablation harnesses), three configurations dominate by mean per-period rank:

- **Paper-sample, all-6-period coverage:** `T+Sym10V3_10s_tiered_ag0`, mean rank $13.33/108$, 5.07–6.06M parameters; top-11 on every M4 period.

- **Paper-sample, paper-faithful alternative:** `NBEATS-IG_30s_ag0`, mean rank $16.0/68$, 38–44M parameters. Use when reproducing Oreshkin et al. [2020] or when zero divergence is required.
- **Sliding, all-6-period coverage:** `T+HaarV3_30s_bd2eq`, mean rank $12.7/112$, 16.25M parameters; smaller alternative `TALG+HaarV3ALG_30s_ag0` (3.66M, mean rank 21.0).

B.9 Reproducibility

Canonical CSVs (`experiments/results/m4/`). Paper-sample (step-paper LR): `comprehensive_m4_paper_sample_results.csv`, `comprehensive_m4_paper_sample_sym10_fills_results.csv`, `tiered_offset_m4_allperiods_paperlr_results.csv`, `m4_hourly_sym10_tiered_offset_paperlr_results.csv`. Paper-sample (plateau LR): `comprehensive_m4_paper_sample_plateau_results.csv`, `tiered_offset_m4_allperiods_results.csv`, `m4_hourly_sym10_tiered_offset_results.csv`. Sliding: `comprehensive_sweep_m4_results.csv`.

Canonical analysis. Meta-leaderboard at `experiments/analysis/analysis_reports/m4_overall_leaderboard_202` regenerable from `experiments/analysis/scripts/m4_overall_leaderboard.py`. The strict health filter `diverged` $\vee$ `smape`$>$`100` $\vee$ `(best_epoch`$=$`0` $\wedge$ `smape`$>$`50)` $\vee$ `smape NaN` is applied to every cell.

Canonical YAML configs (`experiments/configs/`). Paper-sample faithful: `comprehensive_m4_paper_sample.yaml`. Paper-sample plateau: `comprehensive_m4_paper_sample_plateau.yaml`. Sliding: `comprehensive_sweep_m4.yaml`. Tiered: `tiered_offset_m4_allperiods.yaml` (and `m4_hourly_sym10_tiered_offset.yaml` for the dedicated Hourly file).

B.10 Compute resources

All experiments reported in this paper were run on a single workstation: 2$\times$ NVIDIA RTX 5090 (32 GB GDDR7), AMD Ryzen Threadripper 7970X (32C/64T), 128 GB RAM, CUDA 13.0, driver 580.126.09. Each individual run uses one GPU; the two GPUs were used to run independent jobs in parallel from a shared work queue. The M4 paper-sample step budget is fixed at 10^5 gradient steps with batch size 1024 (Appendix A); sliding-window M4 runs at most 75 epochs with patience-20 early stopping (Appendix B.3). The Comprehensive Sweep (112 configs $\times$ 6 M4 periods $\times$ 10 seeds) plus the curated paper-sample (53 configs $\times$ 6 $\times$ 10) and plateau (53 $\times$ 6 $\times$ 10) reruns dominate the budget, with Tourism, Traffic, Weather, and Milk sweeps adding a smaller fraction. The from-scratch SmolLM2-135M-class LLM transfer runs (Appendix G) and the Llama-3.2-1B distillation recovery were the most expensive single jobs but a small fraction of total GPU-hours; only one variant per LLM cell was run, and the SmolLM2 wavelet runs were stopped before the full 3.0 B-token budget as noted in Section 4.5.

C Configuration list

D Additional ablations

E M4 full per-period tables

F Tourism and Milk Out-of-Distribution Results

This appendix collects the Tourism and Milk evidence summarized in Section 4.1 and Section 4.3. Both datasets are small relative to M4 and serve as out-of-distribution checks rather than as primary benchmarks.

F.1 Tourism (Y/Q/M)

Tourism uses a column-oriented data layout (1,311 series, horizons $H \in \{4, 8, 24\}$, $L = 5H$, sMAPE loss) and is evaluated under the same protocol as M4 but with two orders of magnitude fewer series. Table 13 lists the protocol-local winners we observed under the comprehensive-sweep protocol. Tourism-Yearly was also probed by a targeted successive-halving study outside the main 112-config sweep; the best targeted configuration (`TrendWaveletAELG_coif3_eq_bcast_td3_ld16`, sMAPE 20.864) sits 4.2% below the 112-config sweep winner reported here, so the Tourism-Yearly row should not be read as a final SOTA claim.

Table 13: Tourism protocol-local winners (comprehensive-sweep protocol, mean $\pm$ std over surviving seeds).

Period	Metric	Best config	Score	Params	n	N-BEATS ref.
Tourism-Y	sMAPE	`TW 10s td3 db3`	21.773 ± 0.384	2.0M	10	IG-10*: 22.180

The qualitative picture matches the M4 sweep at smaller scale. RootBlock-backed unified `TrendWavelet` blocks lead at the small horizons ($H = 4, 8$); AE-family blocks become more competitive at the larger Tourism-Monthly horizon ($H = 24$). The targeted Yearly study additionally confirmed two M4 defaults on Tourism: `trend_thetas_dim`$= 3$ at $H = 4$ ($p = 0.008$) and `latent_dim`$= 16$ for AELG-backed blocks ($p < 0.02$), both reported in Appendix B.

F.2 Milk

The univariate Milk production series is the small-data stress test (one series, $H = 6$, $L = 5H$, sMAPE loss). The 10-stack winner is `TALG+DB3V3ALG_10s` at sMAPE 1.512 ± 0.572, 1.0M parameters, 10 surviving seeds, against an N-BEATS-IG-10 reference of 1.785. The Milk sweep also reproduces the small-data version of the Generic-stack stability problem reported in Section 4.3: RootBlock and Generic baselines show high run-to-run divergence, while AE-family TrendWavelet models cut divergence to the low single digits and produce the reliable winner above. The convergence-rate breakdown by configuration is recorded in `experiments/results/milk_convergence_10stack/milk_convergence_10stack_results.csv`.

F.3 Backbone-by-regime summary

Across the four out-of-distribution datasets, RootBlock variants are the most reliable choice on small- and medium-horizon problems with many series (M4, Tourism). AE-family bottlenecks lead on long-horizon multivariate Weather and on the small-data Milk regime. The trend-backbone hierarchy paragraph in Appendix B (§B, "Trend backbone hierarchy") gives the matched-seed Wilcoxon evidence for the M4 ordering and the Weather reversal.

G TrendWavelet and AE-MLP on a Transformer LLM

This appendix documents the cross-architecture port summarized in Section 4.5. The full empirical record lives in the sister repository `pellm`; the four primary source reports are listed at the end of this appendix.

G.1 Architecture port

We replace two block types in a Llama-architecture Transformer with the time-series blocks introduced in Section 2. The attention projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`) are replaced by a frozen `TrendWavelet` basis acting on the input-feature axis of the projection weight: 4 polynomial trend components and 28 db3 wavelet components per row, with the basis stored as non-trainable buffers exactly as in Section 2.3. The SwiGLU MLP block is replaced by the same hourglass autoencoder used for `AERootBlock` and `AERootBlockLG` ($\text{hidden} \to \text{hidden}/2 \to d \to \text{hidden}/2 \to \text{hidden}$, with $d = 256$ for the SmolLM2-class runs and $d = 512$ for the Llama-3.2-1B distillation run). Embedding, layer-norm, and residual scaffolding are unchanged.

Table 14: Parameter counts for the SmolLM2-135M-class port. The MLP block accounts for 59% of the baseline parameter count and is the primary compression target. Total compression of the variants is roughly proportional to MLP-block compression.

Variant	Total	MLP block	Attention	Embed + LM head
baseline	134.5M	79.6M	26.5M	28.3M
`ae_mlp`	69.3M	14.4M	26.5M	28.3M
`tw_root_fc_db3_64_tiered_silu`	67.2M	12.3M	26.5M	28.3M

G.2 From-scratch SmolLM2-135M trajectory

We pretrain each variant from scratch on FineWeb-Edu under a fixed 3.0 B-token budget, batch size 1024, mixed-precision AdamW, cosine learning rate schedule with warmup. Only the `baseline` and `ae_mlp` runs reached the full budget; the wavelet variants were stopped at the token counts shown below and the 3.0 B-token endpoint is a calibrated log-linear projection from the late-training slope (calibration anchor: $+0.057$ nats, measured on the two completed runs to remove the cosine-tail flattening bias).

Table 15: Token-matched validation perplexity (in-domain val set, same harness across variants). The wavelet variant closes the gap to baseline monotonically; the gap-to-`ae_mlp` ratio drops from 1.05 at 250 M tokens to under 1.0 by 1.85 B tokens, so the projection ranking reflects an already-realized ordering rather than a slope-fit artifact.

Tokens	baseline	`ae_mlp`	`tw_root_fc_db3_64_tiered_silu`
250 M	50.28	78.98	68.81
1.00 B	26.42	35.10	33.50
1.85 B	–	–	27.64
2.75 B	19.87	26.52	–
3.0 B	19.82	26.45	24.4 (proj.)

Two negative controls confirm the inductive-prior reading. A frozen wavelet basis with no learned reduction in the MLP (`tw_root_db3_64_silu`) collapses at 250 M tokens (PPL 112) and was aborted. A frozen wavelet basis on attention with no learned reduction (`trendwavelet_db3_32`) keeps 17% more parameters than `ae_mlp` but is Pareto-dominated at every matched-token point. The competitive variant (`tw_root_fc_db3_64_tiered_silu` in Table 15) is the one that adds a learned $\mathrm{SiLU}(\mathbf{W}\mathbf{x})$ stage in front of the basis. Macro-average zero-shot accuracy across seven `lm-evaluation-harness` tasks [Gao et al., 2024] at 3.0 B tokens is 38.51% for `baseline`, 35.43% for `ae_mlp`, and 34.95% for the wavelet variant at its 1.85 B-token checkpoint, with LAMBADA carrying most of the regression; per-task numbers are in the source pellm reports.

G.3 Distillation recovery on Llama-3.2-1B

The same AE-LG bottleneck behaves differently when a teacher is available. We swap a single layer-15 SwiGLU MLP in Llama-3.2-1B [Meta AI, 2024] for a `PEBottleneckMLPLG` block at $d = 512$, then run a two-phase pipeline: Phase 1 caches teacher MLP activations and trains the AE block to reconstruct them under MSE; Phase 2 fine-tunes end-to-end with logit-level KD against the unmodified Llama (KD weight $\alpha = 1.0$, temperature 2.0, FineWeb 50k samples).

The recovered model carries roughly 90% MLP-block compression on the swapped layer at 18.81 PPL, -3.3% relative to the unmodified original. Read together with the from-scratch SmolLM2 result, this is consistent with the M4 finding that capacity reductions need a structural prior the optimizer can actually reach: from random init the basis is unreachable, while a teacher (lstsq weight projection or KD logits) provides the rotation the basis is good at decoding.

G.4 Reinvesting saved parameters

The parameter saving is not the headline result on its own; the headline result is what the saved parameters can be redeposited into. On M4 the same trick produces sub-1M-parameter winners

Table 16: Layer-15 AE-LG swap on Llama-3.2-1B with two-phase AE-pretraining and KD fine-tuning. The same block design that pays a $+0.28$-nat plateau from scratch (Section 4.5) recovers below the unmodified original perplexity once the teacher signal is added.

Stage	Validation PPL
Original Llama-3.2-1B (no swap)	19.45
Baseline (PE layer inserted, no KD, no AE pretrain)	42.72
After AE pretrain (teacher-activation reconstruction)	25.34
After KD fine-tune	18.81

like `TWAELG_10s_ld32_db3` (0.48M params, top-5 on Yearly, Daily, and Hourly under the paper-sample protocol; Table 6). On NHiTS the same blocks transfer cleanly to the multi-rate pooled backbone (Section 4.4). On the Transformer the 50% block-budget saving creates room for richer composite bases (`tw_root_fc_post_fc` with a learned post-basis mixer projects 24.1 PPL at 43.5% compression, slightly better than the 50%-compression headline variant), tiered per-layer offset schedules, and depth, without exceeding the original parameter budget.

G.5 Caveats and reproducibility

Single seed per variant in the SmolLM2-class runs; cross-seed standard deviation is unmeasured. Recipe drift between the legacy 2-GPU DDP runs (`baseline`, `ae_mlp`) and the 1-GPU runs (wavelet variants) is absorbed by the calibration anchor described above; reported gaps are an order of magnitude larger than any plausible drift. Only `baseline` and `ae_mlp` reached the full 3.0 B-token budget; wavelet rows in Table 15 are partial-checkpoint values, with the 3.0 B endpoint a projection. Source reports with the full projection methodology, calibration arithmetic, per-task lm-eval breakdowns, AE-pretraining init hierarchy, and fc2/fc3 SVD rule are at `pellm/evals/smol_replacement_paper/preliminary_feasibility_report.md`, `analysis_tw_root_fc_vs_ae_mlp_vs_baseline.md`, `analysis_ae_pretraining.md`, and `experiments/analysis/analysis_reports/pellm_validation_and_theory_revision.md`.

H Limitations

The architectural sweep is centered on the M4 benchmark; while we include Weather and Traffic as primary out-of-distribution checks (with Tourism and Milk in Appendix F), the breadth of structured-prior conclusions outside M4 is bounded by the size of those auxiliary sweeps. The compute budget supported 10 seeds per cell; some 1–2 sigma effects therefore retain residual seed sensitivity. Pure VAE-family blocks were observed to diverge under iteration-based paper sampling and are excluded *a priori*; we do not claim VAE backbones are unworkable in general, only that they were not stabilized within this study. Finally, all reported gains assume a single architecture is trained per dataset/period; ensembling, common in published M4 results, is intentionally outside scope to keep the parameter-count comparison clean.

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